



UNIFIED ACADEMIC HUB

ICTWS-II Mini Project Report

A centralized web platform for academic resources, assignments, attendance, marks, announcements, and communication

Application Name	Unified Academic Hub
Team Name	Power Boys (Group 1)
Course	ICTWS-II: App and Web Development Laboratory
Institution	Indian Institute of Information Technology, Surat
Academic Year	2025-2026

Submitted By

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Submitted To

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May 2026

Certificate

This is to certify that the project entitled Unified Academic Hub, submitted by Team Power Boys (Group 1), has been satisfactorily completed as part of the ICTWS-II Mini Project / App and Web Development Laboratory requirements for the academic year 2025-2026.

The project represents original work carried out by the students listed below. The system demonstrates a functional academic hub for teachers and students, with modules for authentication, subject management, course resources, assignments, announcements, attendance, marks, and messaging. External resources and documentation used during development have been acknowledged in the bibliography.

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This project has been carried out in partial fulfilment of the laboratory component of ICTWS-II and represents original work completed by the above-named group. All external references and resources used in the development of this project have been duly acknowledged.

Faculty Signature	_____
Date	_____29/04/2026_____
Department Seal	_____

Acknowledgement

We, the members of Team Power Boys (Group 1), express our sincere gratitude to the ICTWS-II faculty and the Indian Institute of Information Technology, Surat, for providing the opportunity to design and develop this mini project. The course introduced us to practical web development concepts and helped us convert classroom learning into a usable academic platform.

We are thankful for the guidance, feedback, and structured learning environment provided during the project development cycle. The project helped us strengthen our understanding of HTML, CSS, JavaScript, DOM manipulation, localStorage, responsive interface design, and client-side application logic.

We also acknowledge the documentation and learning resources from MDN Web Docs, W3Schools, Android Studio documentation, Netlify, GitHub, and the broader developer community. Their references helped us implement and refine the modules of Unified Academic Hub.

- Dhiwyansh Parmar, Aryan Gupta, Rohan Kaushik, Nihal Somani

Team - Power Boys (Group 1),

IIIT Surat | April 2026

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1. Introduction and Abstract

1.1 Introduction

Unified Academic Hub (UAH) is a web-based academic management platform created to simplify day-to-day communication and resource management between teachers and students. In a typical academic environment, course materials, announcements, assignments, attendance records, marks, and messages are often spread across different tools or managed manually. UAH brings these workflows into one centralized interface.

The project is built as a multi-page front-end application using HTML, CSS, and JavaScript. It includes separate screens for login/signup, dashboard, subject-level modules, and messaging. It supports two user roles, student and teacher, and changes available actions based on the logged-in user's role. Teachers can create subjects, post resources, create assignments, mark attendance, assign marks, and communicate with students. Students can join subjects, access materials, submit assignments, view attendance and marks, and message teachers.

1.2 Abstract

Unified Academic Hub is a client-side academic portal designed for IIIT Surat-style classroom workflows. The application provides role-based authentication, subject enrollment, course material sharing, assignment creation and submission, announcement posting, attendance tracking, marks management, and direct messaging in a single responsive dashboard. Browser localStorage is used to persist users, subjects, enrollments, resources, submissions, messages, attendance, and marks without requiring a backend server in the current prototype.

The project demonstrates practical knowledge of semantic HTML structure, modular CSS design, JavaScript event handling, DOM manipulation, FileReader-based uploads, JSON-based local data storage, responsive layouts, dark/light theme handling, and CRUD operations. It is designed as a functional proof of concept that can later be extended with a secure backend, database, cloud synchronization, analytics, and native mobile application support.

2. Problem Statement and Proposed Solution

2.1 Problem Statement

Students and teachers often face difficulty when academic activities are managed through scattered tools. Notes may be shared through separate drives, assignments through messages, attendance through manual registers, and marks through individual spreadsheets. This fragmentation increases confusion, delays communication, and makes it difficult for students to track their academic responsibilities from one place.

- Course materials and notices are scattered across different platforms.
- Attendance and marks are often handled manually, creating extra workload.
- Teacher-student communication is not always linked to the relevant subject.
- Students lack a single dashboard for subjects, assignments, messages, and academic status.
- Teachers need a simple interface to manage class records and updates.

2.2 Proposed Solution

UAH solves this problem by providing one academic hub where teachers and students can manage subject-specific activities. Instead of switching between multiple tools, users can sign in, view their subjects, access each subject's tabs, and complete academic tasks from a consistent interface.

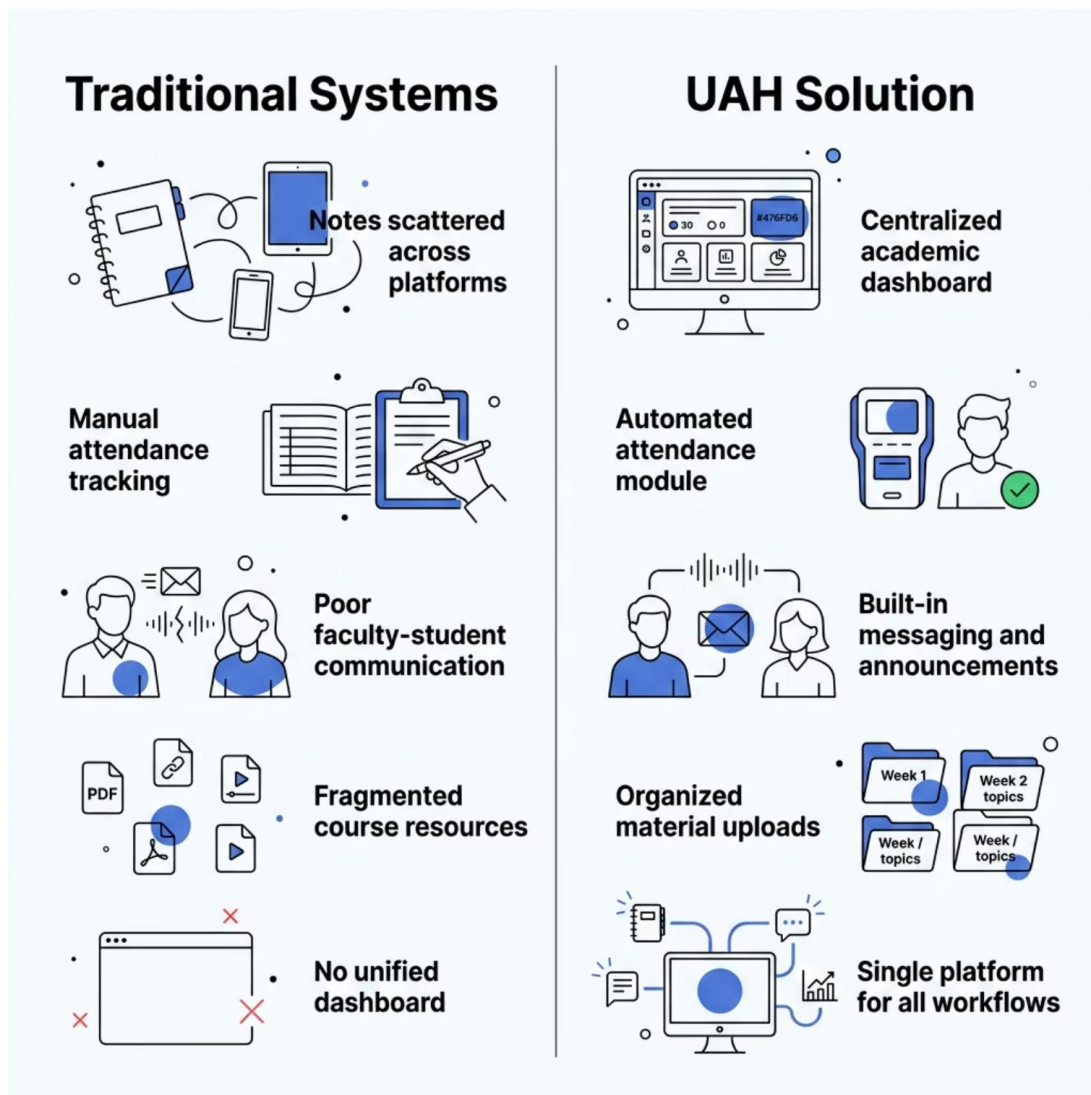


Figure 1: Traditional academic systems compared with the UAH solution.

Traditional System Issue	UAH Solution
Scattered notes and files	Centralized material upload and download module
Manual attendance records	Digital attendance tracking with percentage view
Assignment confusion	Subject-wise assignment creation, deadline display, and student submission
Disconnected communication	Built-in subject messaging and inbox conversations
No unified overview	Dashboard with subjects, assignments, students, and unread message indicators

3. Platform and Tools Used

The submitted implementation is a browser-based web application. The presentation also lists Android Studio, Java, and XML as project tooling references; however, the source files provided for this report are the web implementation files.

Technology / Tool	Role in Project	Platform
HTML5	Defines the structure of login, dashboard, subject, and inbox pages.	Web
CSS3	Provides layout, color system, responsive behavior, cards, tabs, modals, and theme styling.	Web
JavaScript ES6	Implements authentication, CRUD operations, dashboard rendering, messaging, validation, and state handling.	Web
DOM API	Updates views dynamically based on role, subject selection, and stored data.	Web
localStorage API	Stores users, subjects, enrollments, materials, assignments, submissions, attendance, marks, and messages.	Web
FileReader API	Supports local file upload previews/storage for materials, assignments, and submissions.	Web
Visual Studio Code	Used as the development editor for the web files.	Development
GitHub	Used as a repository and source sharing platform.	Development
Netlify	Listed as deployment and hosting platform in project references.	Deployment
Gamma	Used for project presentation design.	Documentation
Android Studio / Java / XML	Listed in the PPT as additional tooling context for app development exploration.	Reference

4. Objectives and Workflow

4.1 Objectives

- To provide a single academic dashboard for students and teachers.
- To simplify subject creation, subject joining, and subject-wise navigation.
- To centralize course materials, links, files, and learning resources.
- To digitize assignment creation, file attachment, student submission, and submission tracking.
- To provide teachers with attendance and marks management tools.
- To give students quick visibility into attendance percentage, marks, assignments, and messages.
- To support direct communication through subject messages and an inbox module.
- To demonstrate front-end application development using HTML, CSS, JavaScript, and localStorage.

4.2 Workflow Diagram

The user journey begins with the landing page and role-based login. After authentication, the dashboard becomes the central navigation point. Users can then move into subject management and access modules such as materials, assignments, announcements, attendance, marks, and messages.

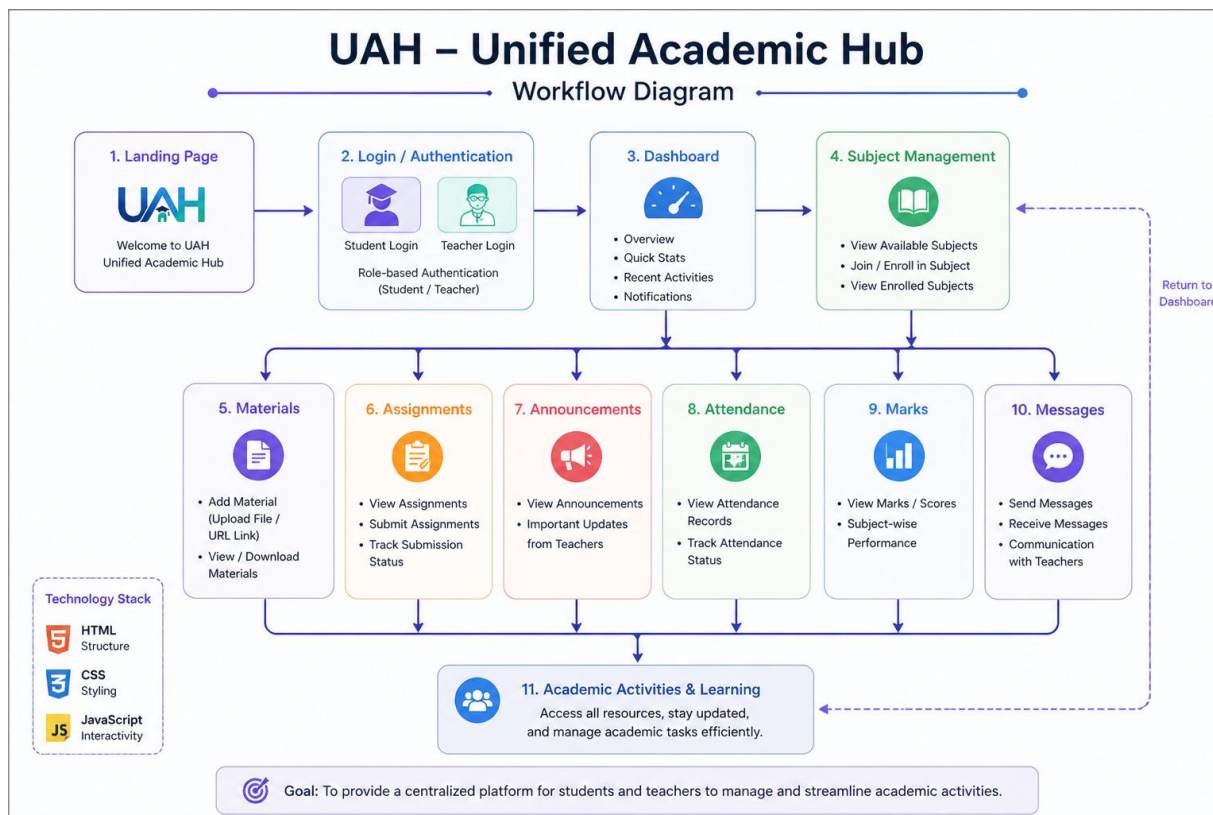


Figure 2: UAH workflow diagram showing the main user journey and academic modules.

5. System Design and Data Architecture

5.1 Page Structure

File	Purpose
index.html	Landing, login, and signup interface with student/teacher role toggle.
dashboard.html	Role-based overview screen for subject creation, subject joining, statistics, and navigation.
subject.html	Main subject workspace containing materials, assignments, announcements, attendance, marks, and messages tabs.
inbox.html	Conversation-based messaging interface between connected teachers and students.
script.js	Application logic for authentication, data storage, CRUD operations, rendering, validation, and messaging.
styles.css	Complete visual design system with responsive layout, cards, tabs,

	tables, modals, dark mode, and mobile support.
assets/logo.png	Project branding used across the application and report.

5.2 Local Storage Data Model

UAH uses browser localStorage as the persistence layer for the prototype. Each data category is stored as a JSON array or object and is retrieved through helper functions in script.js.

Storage Key	Data Stored	Usage
users	Registered users	Login, signup, role checks, names, emails, roll numbers
currentUser	Active user object	Session management and role-based UI
subjects	Created subjects	Teacher-owned subjects and student subject lists
enrollments	Student-subject mapping	Join subject flow and enrolled student lookup
materials	Links/files for resources	Course material display and download
assignments	Assignment records	Teacher assignment creation and deadline display
submissions	Student submissions	Submission history and teacher review
announcements	Subject announcements	Teacher posts and student updates
attendance	Date-wise presence records	Attendance percentage and records
marks	Exam/test marks	Student marks view and teacher marks table
messages	Direct and subject messages	Inbox threads and unread indicators
theme	Light/dark preference	Persistent UI theme selection

6. Working of the Project

6.1 Authentication and Role Selection

The application begins with a landing screen followed by login/signup forms. Users select either Student or Teacher mode. Email validation requires the institutional domain @iiitsurat.ac.in, passwords must be at least six characters, students provide a roll number, and teachers must enter the teacher passkey. After successful login, the active user is saved in currentUser and redirected to the dashboard.

6.2 Dashboard

The dashboard changes according to role. Teachers see their created subjects, total enrolled students, and unread messages. Students see enrolled subjects, assignment count, and unread messages. Teachers can create subjects by entering a name and code, while students can join available subjects.

6.3 Subject Workspace

Each subject opens in subject.html and contains tabbed modules. The subject title and code are loaded from localStorage using the subject id passed in the URL. Teacher-only actions are displayed only when the logged-in user has the teacher role.

6.4 Materials Module

Teachers can add course materials either as URL links or uploaded files. Materials are categorized as notes, PDF, lecture, or other. Students can view links or download uploaded files from the subject page.

6.5 Assignments and Submissions

Teachers create assignments with title, description, optional attachment, deadline, and grade points. Students can submit files with optional notes. Teachers can see the submissions linked to each assignment, and assignment deletion also removes related submissions.

6.6 Announcements

Teachers post announcements with title and content. Announcements are displayed in reverse chronological order so the latest updates are visible first. This module reduces dependence on separate messaging channels for subject notices.

6.7 Attendance

Teachers mark attendance date-wise for enrolled students using checkboxes. The system stores present/absent status for every student and date. Students can view their own attendance percentage and receive a shortage warning when attendance falls below 75 percent.

6.8 Marks

Teachers assign marks exam-wise for enrolled students. Marks can be edited or deleted by exam name. Students view their own exam marks, total marks, and percentage badges.

6.9 Messaging and Inbox

The project includes subject messages as well as a global inbox. Teachers can message enrolled students, and students can message subject teachers. The inbox groups conversations, shows unread indicators, and marks messages as read when a chat is opened.

6.10 Theme and Responsive Interface

The stylesheet defines a reusable design system with CSS variables, cards, buttons, tabs, tables, modals, badges, toasts, and responsive behavior. A theme toggle stores the user's dark/light preference in localStorage.

7. Code Explanation

The implementation uses a simple modular structure. HTML files define the screen layouts, styles.css provides the complete UI system, and script.js contains the application behavior. The JavaScript file is divided into clear functional sections: utilities, toast notifications, theme handling, authentication, sidebar, dashboard, subject CRUD, subject page modules, and inbox.

Module	Key Functions / Elements	Explanation
Utilities	uuid, getLS, setLS, fmtDate, fmtTime	Reusable helpers for IDs, JSON storage, and date formatting.
Authentication	handleLogin, handleSignup, requireAuth, logout	Manages login, signup validation, role matching, and session control.
Dashboard	initDashboard, renderSubjectsGrid	Builds role-based statistics and subject cards dynamically.
Subject CRUD	createSubject, deleteSubject, openJoinSubjectModal, joinSubject	Allows teachers to create/delete subjects and students to join available subjects.
Materials	loadMaterials, addMaterial, deleteMaterial	Stores and displays links or uploaded resource files.
Assignments	loadAssignments, addAssignment, submitAssignment	Handles assignment creation, deadlines, attachments, and student submissions.
Announcements	loadAnnouncements, addAnnouncement	Displays teacher-posted notices in a subject.
Attendance	loadAttendance, saveAttendance, viewStudentAttendance	Stores date-wise attendance and computes percentages.
Marks	loadMarks, saveMarks, editExamMarks	Stores exam marks and shows teacher/student views.
Messages	loadSubjectMessages, sendSubjectMessage, loadConversations, openChat	Supports subject-level messaging and inbox conversations.

7.1 Validation and User Safety

- Only institutional email addresses ending with @iiitsurat.ac.in are accepted.
- Passwords require a minimum length of six characters.
- Teacher signup is protected through a passkey.
- Student signup requires a roll number.
- Delete operations use confirmation prompts before removing subject, assignment, or module data.

8. Results and Screenshots

The following screenshots show the implemented interface and core workflows. The mobile screenshots demonstrate responsive behavior, while the desktop subject screenshot shows the tab-based academic workspace.

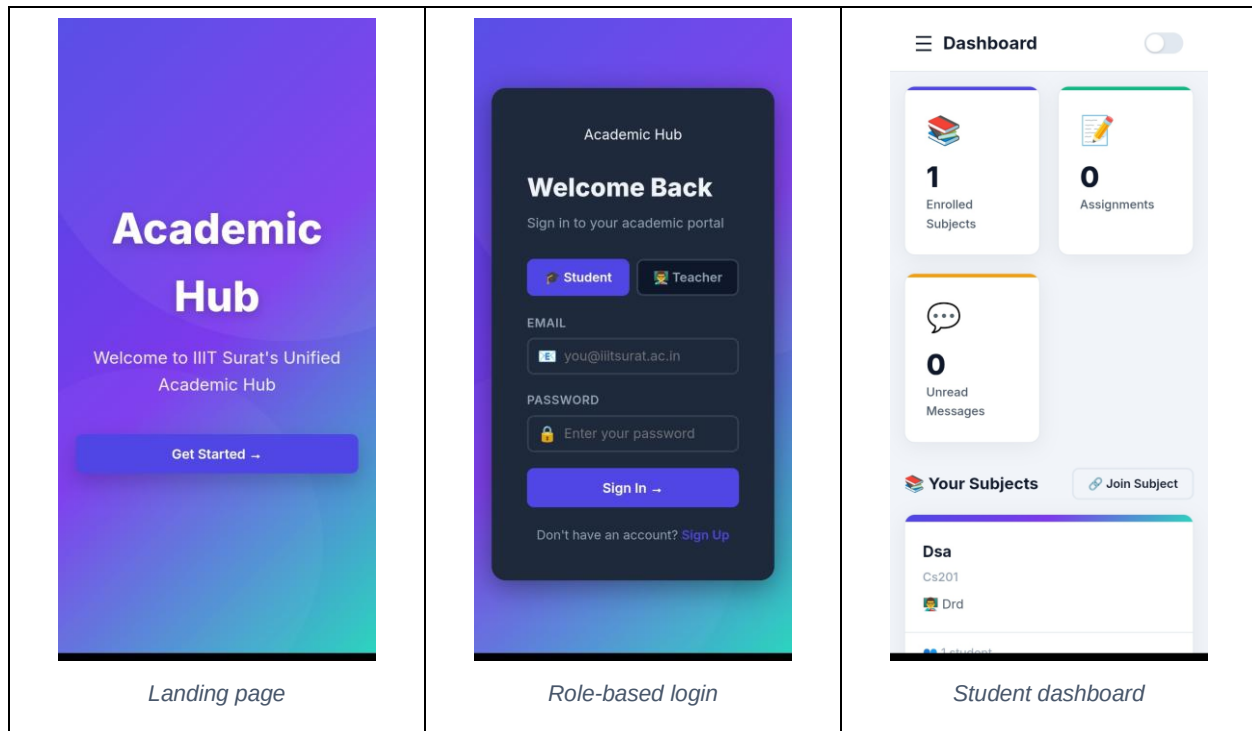


Figure 3: Responsive UI screens from Unified Academic Hub.

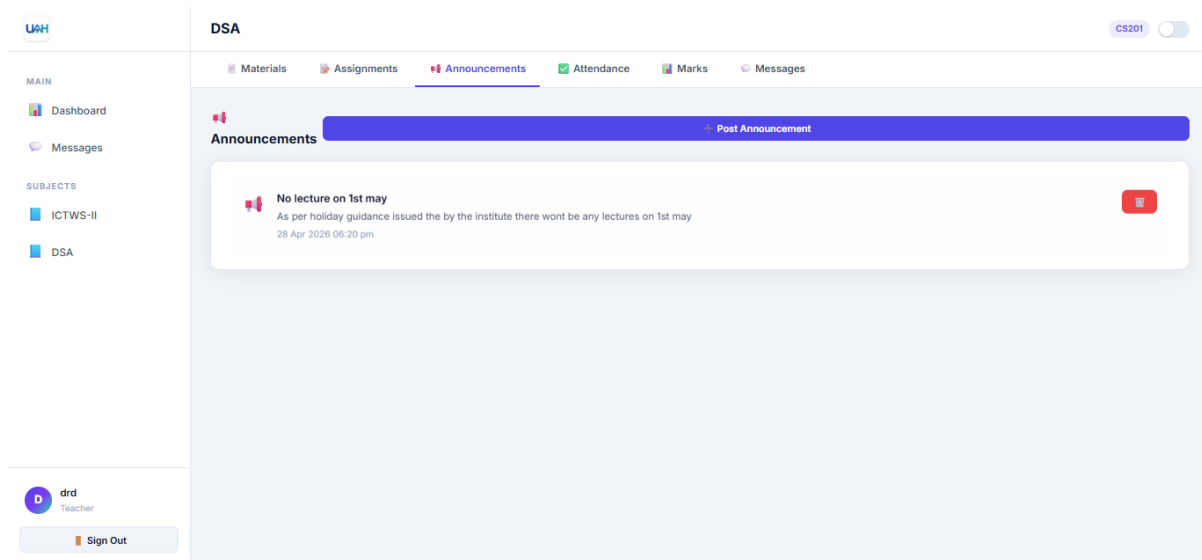


Figure 4: Desktop subject page with tabbed modules and announcements.

Unified Academic Hub - Project Report

UAH

MAIN

Dashboard

Messages

SUBJECTS

WEB DEV

ROHAN

Teacher

Sign Out

WEB DEV

CS201

Materials

Assignments

Announcements

Attendance

Marks

Messages

Course

Materials

Add Material

LEC 1

IMPORTANT

Download: TEAM NAME (4).pdf

1 May 2026 07:30 pm

UAH

MAIN

Dashboard

Messages

SUBJECTS

WEB DEV

ROHAN

Teacher

Sign Out

WEB DEV

CS201

Materials

Assignments

Announcements

Attendance

Marks

Messages

Assignments

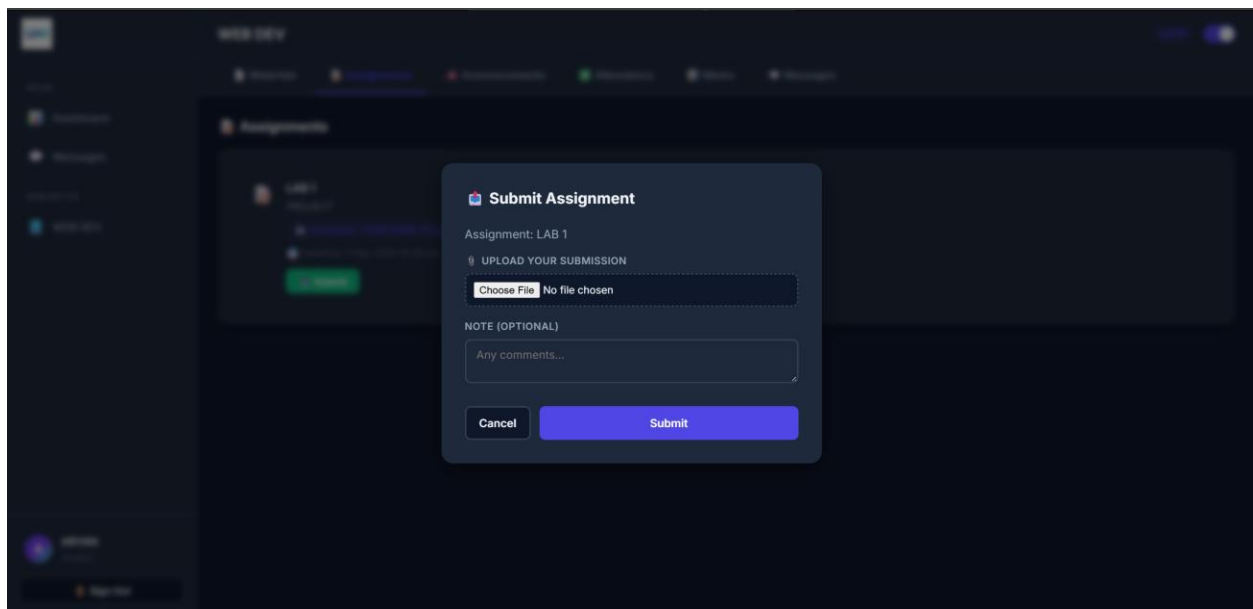
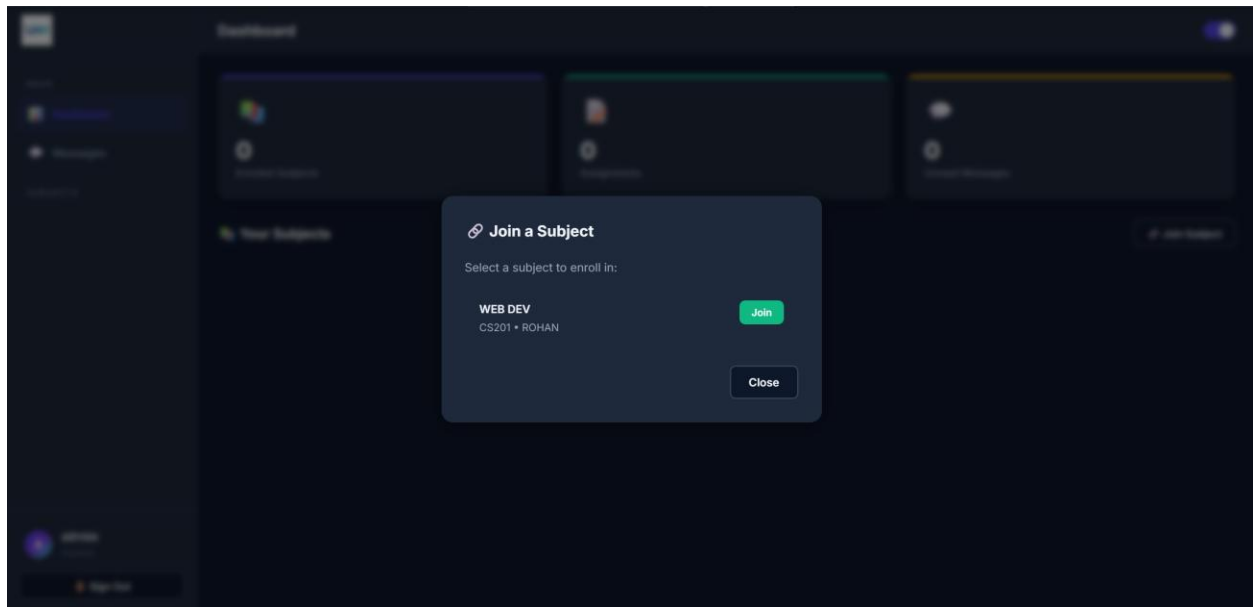
Create Assignment

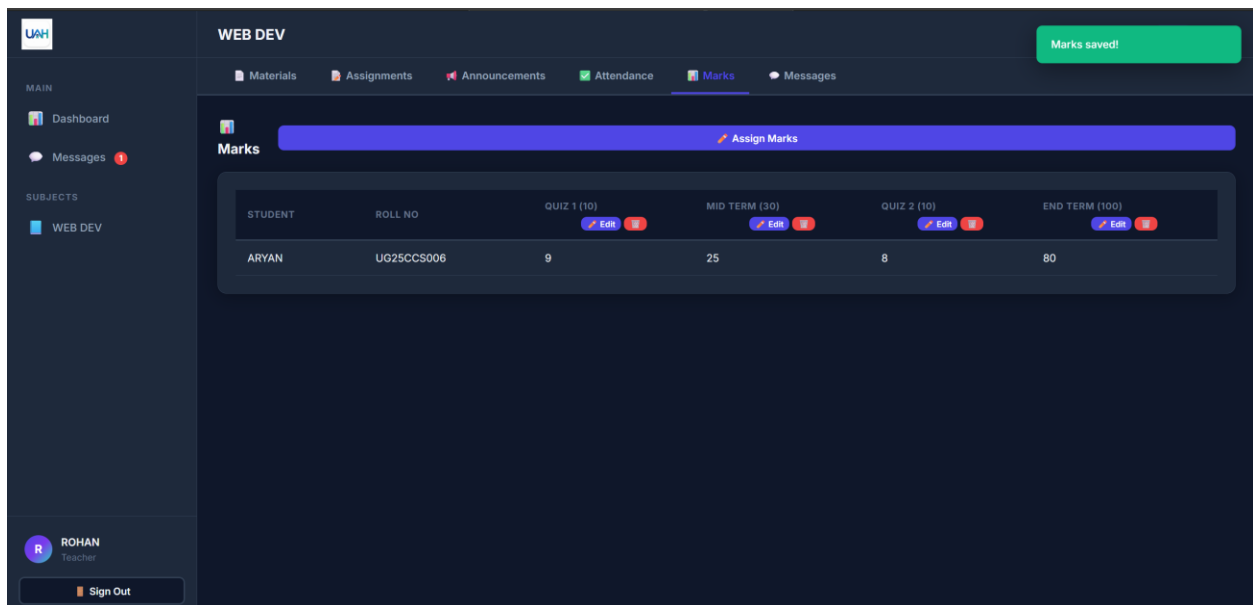
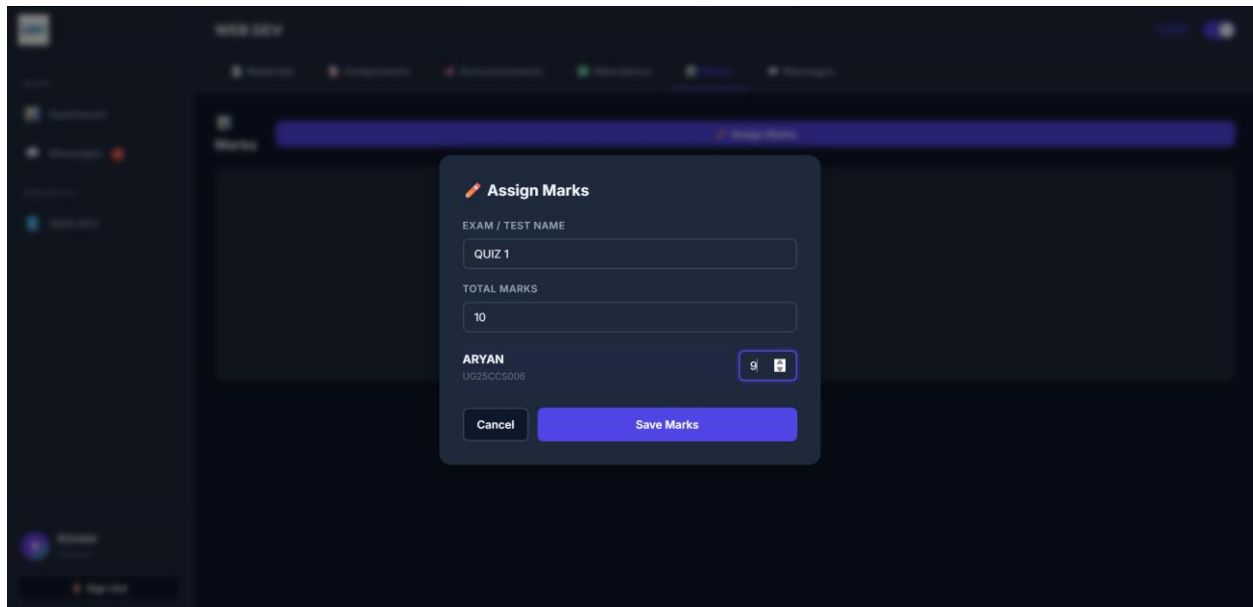
LAB 1

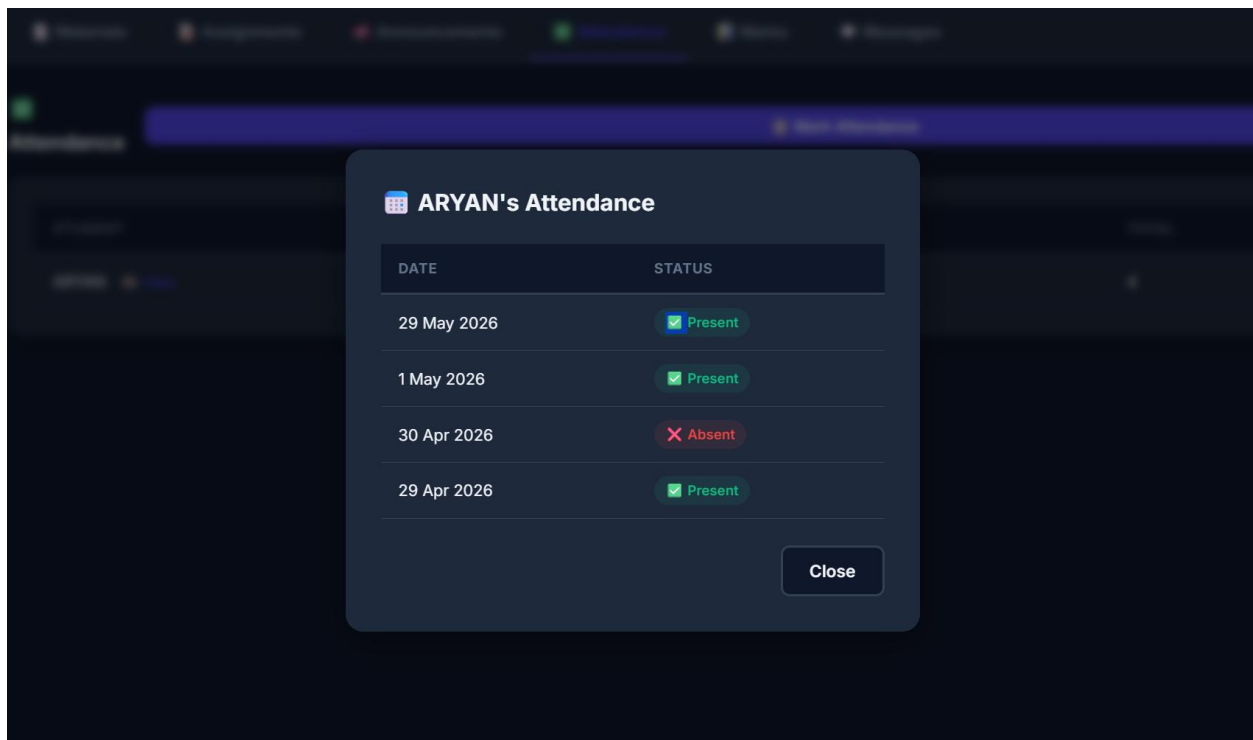
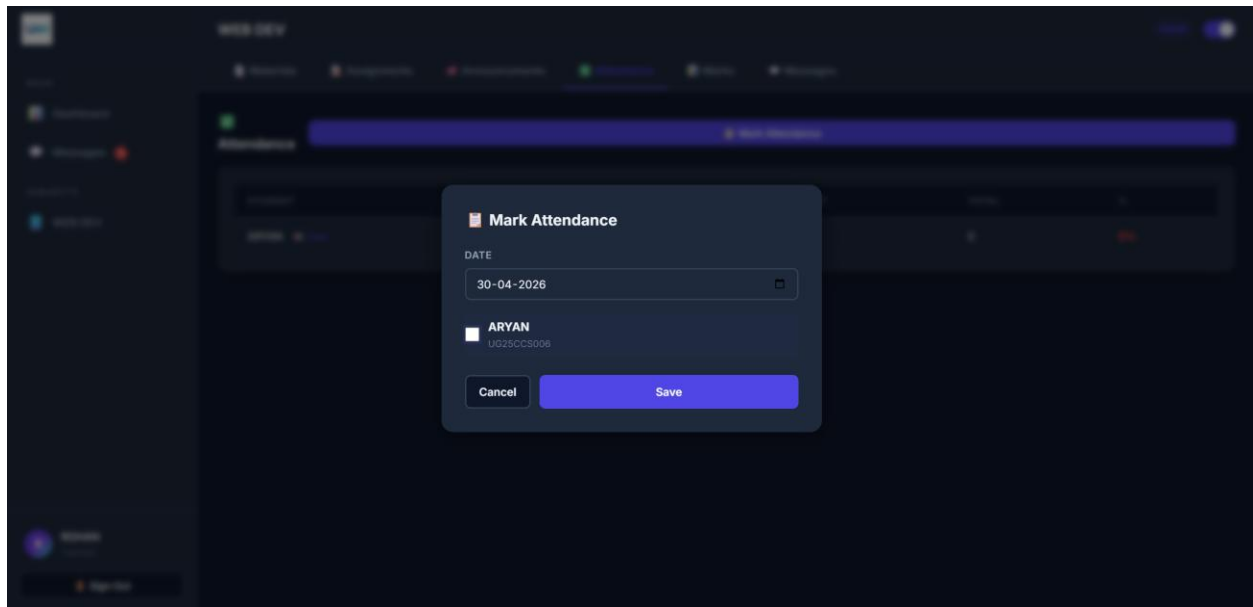
PROJECT

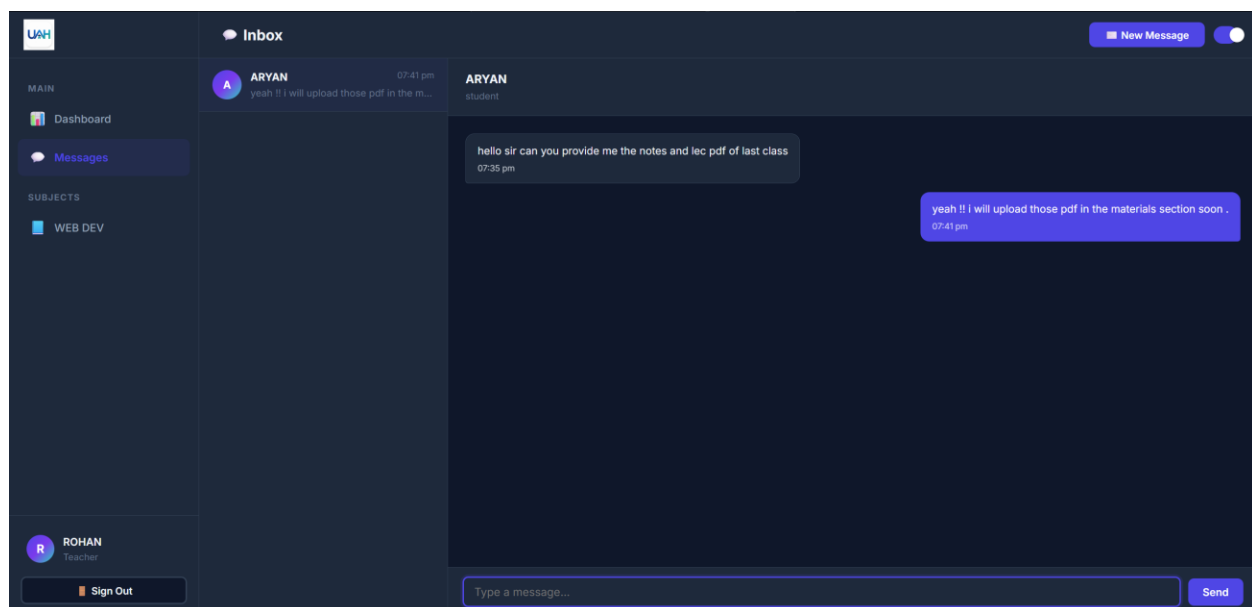
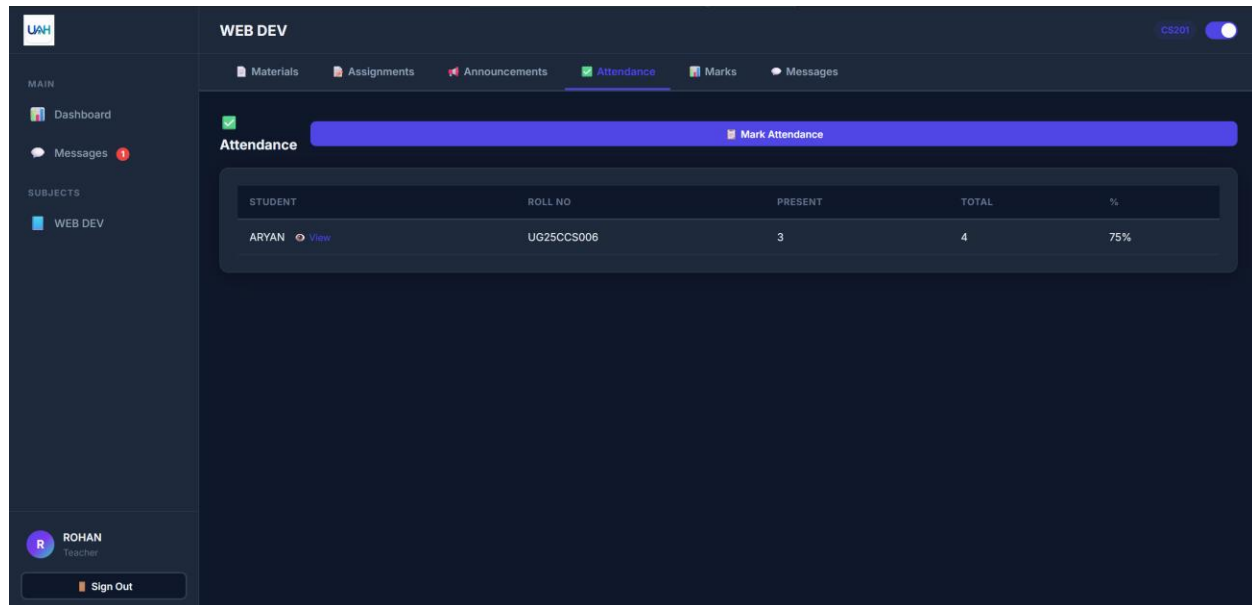
Download: TEAM NAME (2).pdf

Deadline: 1 May 2026 12:59 pm **Overdue** 100 Points









8.1 Functional Output Summary

Feature	Observed Result
Login and Signup	Role-based login/signup forms with validation for email domain, password length, roll number, and teacher passkey.
Dashboard	Displays statistics and subject cards based on teacher or student role.
Subject Management	Teachers create subjects; students join available subjects.
Materials	Supports URL links and uploaded files for learning resources.
Assignments	Teachers create assignments; students submit files and notes.
Announcements	Teachers post subject notices for students.
Attendance	Teachers mark attendance; students view percentage and shortage

	warning.
Marks	Teachers assign exam marks; students view their scores and percentages.
Messaging	Subject messages and inbox conversations with unread badges.

9. Benefits, Limitations, and Future Scope

9.1 Benefits for Students

- Students get a single place to view subjects, assignments, materials, attendance, marks, and messages.
- Subject-wise organization reduces confusion and saves time during academic work.
- Attendance and marks visibility helps students track performance and take timely action.
- Direct messaging improves communication with teachers.

9.2 Benefits for Teachers

- Teachers can create subjects and manage student academic records from one dashboard.
- Announcements, materials, assignments, attendance, and marks are managed subject-wise.
- Submission tracking and student messaging reduce manual coordination effort.

9.3 Current Limitations

- Data is stored in browser localStorage, so it is device-specific in the current prototype.
- The current implementation does not include backend authentication or database security.
- File upload storage is suitable for demonstration but should be moved to cloud storage in production.
- Role verification is client-side and should be strengthened before real institutional deployment.

9.4 Future Scope

- Cloud backend with database support for secure cross-device synchronization.
- OAuth or institute single sign-on for safer authentication.
- AI study assistant for smart summaries, recommendations, and doubt support.
- Smart attendance analytics with trends and predictive shortage alerts.
- Native Android and iOS applications for mobile-first access.
- Integrated video lecture upload, playback, and subject-wise archive.
- Performance dashboards with charts for marks, attendance, submissions, and engagement.
- Push/email notifications for announcements, deadlines, attendance shortage, and unread messages.

10. Conclusion

Unified Academic Hub successfully demonstrates how a compact front-end application can simplify academic management for both students and teachers. By combining subject management, resources, assignments, submissions, announcements, attendance, marks, and messaging in one interface, the project addresses a real workflow problem found in many academic settings.

The project fulfills the objectives of the ICTWS-II mini project by applying HTML, CSS, and JavaScript to build a functional, role-aware, responsive web application. The current version works as an offline-capable prototype using localStorage. Its structure is also suitable for future expansion into a production-grade system with backend storage, secure authentication, cloud sync, analytics, and mobile application support.

11. Bibliography and References

- HTML5 Documentation - <https://developer.mozilla.org>
- CSS3 Documentation - <https://developer.mozilla.org>
- JavaScript Documentation - <https://developer.mozilla.org>
- MDN Web Docs: localStorage API - <https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage>
- MDN Web Docs: Document Object Model - https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model
- Google Android Studio Official Documentation - <https://developer.android.com/studio>
- W3Schools HTML, CSS, and JavaScript References - <https://www.w3schools.com>
- Netlify - Website deployment and hosting - <https://www.netlify.com>
- Gamma - Presentation design tool - <https://gamma.app>
- Project GitHub Repository - <https://github.com/ARYAN-GUPTA-105/UNIFIED-ACADEMIC-HUB---ICTWS-PROJECT>
- Project screenshots drive folder - <https://drive.google.com/drive/u/0/folders/10J0pUZv4BzUciZ4D7RuJh09gDPNdORUV>